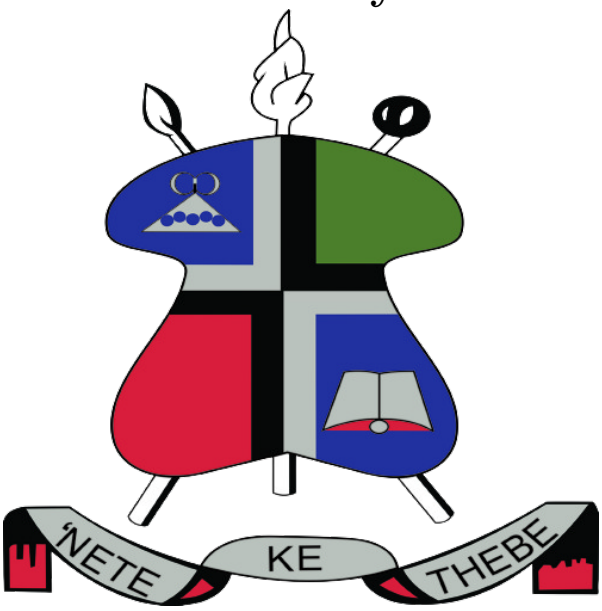
National University of Lesotho



Exploratory Data Analysis

Detection and Classification of Lung Cancer using Deep Learning

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1 Step 1: Understanding the Problem and the Data

1.1 1. Problem Definition

The main goal of this project is to build a deep learning model that classifies CT scans as:
1. Normal: standard lung structures that function normally without abnormal buildup, mass or scarring.
Imaging: Shows a clear air-filled lung parenchyma on a CT scan.
2. Benign: non-cancerous tissue that grows slowly and does not metastasize to other organs.
Imaging: smooth or round shapes, well defined edges and a higher chance of internal calcification.
3. Malignant: uncontrolled growth of abnormal or mutated cells that can invade surrounding lung tissue and spread to distant organs.
Imaging: irregular, lobulated or spiculated margins and thick-walled cavities.

The project shall help distinguish healthy lung tissue from abnormal tissue and identify cancerous lesions.

1.2 2. Dataset overview

The dataset is the IQ-OTH/NCCD lung cancer dataset with this structure:

Table 1: Summary of the IQ-OTH/NCCD lung cancer CT dataset.

Attribute	**Description**
Dataset name	IQ-OTH/NCCD
Source institution	Iraq-Oncology Teaching Hospital / National Center for Cancer Diseases
Year collected	2019
Availability	Publicly available on Kaggle
Data type	CT scan slices (JPEG images)
Total images	1097
Total cases	110
Privacy	All images de-identified before analysis

The dataset images were marked by oncologists and radiologists from the two centers and originate from a Siemens SOMATOM scanner with slice thickness of 1 mm.

Table 2: Class distribution of cases in the IQ-OTH/NCCD dataset.

Class	**Cases**	**Images**	**Share of Cases(%)**
Normal	55	416	50.0
Malignant	40	561	36.4
Benign	15	120	13.6
Total	**110**	**1097**	**100.0**

1.3 Nature of the data

The dataset images are 2D CT slice that are stored as JPEG files. Each image represents a chest CT slide, and there are multiple slices per patient. Therefore, there is patient-level grouping and there is a chance of near-duplicate slices from adjacent CT slices.